



Exercises session 3

Polynomial time complexity II

1 Definition

Definition 1.1 (Vertex colouring) A vertex colouring of a graph $G = (V, E)$ with k colours is a function

$$c : V(G) \rightarrow \{1, \dots, k\}$$

such that adjacent nodes have different colours, that is

$$(u, v) \in E \rightarrow c(u) \neq c(v)$$

Definition 1.2 (Conjunctive normal form) A propositional logic formula φ is in conjunctive normal form (CNF) if

$$\varphi = C_1 \wedge \dots \wedge C_m$$

where each C_i is a disjunction of literals

$$C_i = (\ell_{i1} \vee \dots \vee \ell_{ik})$$

and a literal is either a variable or the negation of a variable.

We also say that a formula is in k -CNF if it is in CNF and each clause has at most k literals.

2 Session exercises

1. Consider the following formula:

$$\varphi = (\neg x_1 \vee x_2) \wedge (\neg x_2 \vee x_3) \wedge (\neg x_3 \vee x_4) \wedge (\neg x_4 \vee x_1) \wedge (x_1 \vee x_5) \wedge (\neg x_5)$$

- (a) Determine if the formula is satisfiable using the approach seen for the 2-SAT problem
- (b) Repeat previous step, but now using the approach seen for HORN-SAT
- (c) Compare the running times of both methods.

2. Consider the following problem:

$$2\text{-COL} = \{G : G \text{ has a 2-colouring}\}$$

- Design an algorithm that decides if a graph has a 2-colouring
 - Relate this problem to 2-SAT and conclude that 2-COL is in PTIME.
3. We saw in classes that Horn-SAT is solvable in polynomial time. However, sometimes not all clauses can be satisfied. In such cases, we seek an assignment that satisfies the maximum number of clauses—this is the **Max-Horn-SAT** problem.
- Given the following Horn formula:

$$\varphi = (\neg x_1 \vee x_2) \wedge (\neg x_2 \vee x_3) \wedge (\neg x_3 \vee x_4) \wedge (\neg x_4 \vee x_1) \wedge (\neg x_3) \wedge (x_4)$$

- (a) Determine if a satisfying assignment exists.
- (b) If the formula is unsatisfiable, determine the **maximum number of clauses** that can be satisfied.
- (c) Provide an example of an assignment that **maximizes satisfied clauses**.
- (d) Can you think of an efficient algorithm for solving Max-Horn-SAT?

1. Consider the following formula:

$$\varphi = (\neg x_1 \vee x_2) \wedge (\neg x_2 \vee x_3) \wedge (\neg x_3 \vee x_4) \wedge (\neg x_4 \vee x_1) \wedge (x_1 \vee x_5) \wedge (\neg x_5)$$

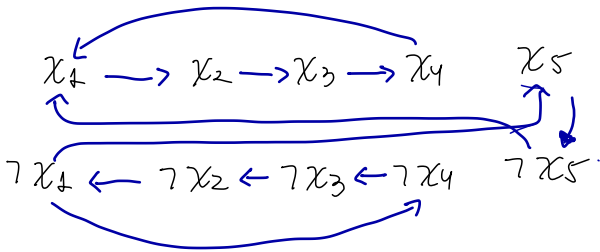
$x_4 \rightarrow x_1$ $\neg x_4 \rightarrow x_5$
 $\neg x_1 \rightarrow \neg x_4$ $\neg x_5 \rightarrow x_1$
 $\neg x_5 \vee \text{False}$
 $x_5 \rightarrow \text{False}$
 $\text{True} \rightarrow \neg x_5$

- Determine if the formula is satisfiable using the approach seen for the 2-SAT problem
- Repeat previous step, but now using the approach seen for HORN-SAT
- Compare the running times of both methods.

(a) We use an **incidence graph**; each clause $a \vee b$ corresponds to 2 edges:

$$\neg a \rightarrow b$$

$$\neg b \rightarrow a$$



No variable and its negation are in the same SCC.

SCC $\{x_1, x_2, x_3, x_4\}$

$\{ \neg x_1, \neg x_2, \neg x_3, \neg x_4 \}$

$\{x_5\}$

$\{ \neg x_5 \}$

As we have $x_5 \rightarrow \neg x_5$; we know that $x_5 = \text{False}$;

so an assignment will be $x_1, x_2, x_3, x_4 = \text{True}$
 $x_5 = \text{False}$.

(b) We use the unit propagation technique.

$$\varphi = (\overset{C_1}{\neg x_1 \vee x_2}) \wedge (\overset{C_2}{\neg x_2 \vee x_3}) \wedge (\overset{C_3}{\neg x_3 \vee x_4}) \wedge (\overset{C_4}{\neg x_4 \vee x_1}) \wedge (\overset{C_5}{x_1 \vee x_5}) \wedge (\overset{C_6}{\neg x_5})$$

① start with $\neg x_5$

$$x_5 = \text{False}$$

② then in C_5 , $x_1 \vee \text{False}$

$$\Rightarrow x_1 = \text{True}$$

③ In C_4 , $\neg x_1 \vee x_2 \Rightarrow \text{False} \vee x_2$

$$\Rightarrow x_2 = \text{True}$$

④ In C_2 , $\neg x_2 \vee x_3 \Rightarrow \text{False} \vee x_3$

$$\Rightarrow x_3 = \text{True}$$

⑤ In C_3 , $\neg x_3 \vee x_4 \Rightarrow \text{False} \vee x_4$

$$\Rightarrow x_4 = \text{True}$$

⑥ Finally, in C_1 $\neg x_4 \vee x_1$

$$\neg x_4 = \text{False}$$

$$x_1 = \text{True}$$

①

No contradictions, and we have

$$x_1, x_2, x_3, x_4 = \text{True}$$

$$x_5 = \text{False}.$$

as a satisfying assignment.

(c) Comparison UNIT PROPAGATION & INCIDENCE GRAPH.

Formula ϕ in CNF ; m clauses, n variables

► INCIDENCE GRAPH

- graph construction :
 $2m$ edges $\rightarrow \Theta(m)$
- Find strongly connected components
 $\hookrightarrow$ can be done in
 $\Theta(V + E)$
 $\therefore \Theta(m + m)$
- check if x_i and $\neg x_i$ are in the same SCC
 $\Theta(m)$

$\therefore$ overall time : $\Theta(m + m)$

► UNIT PROPAGATION

- a unit clause $\{x_i\}$ is processed in $\Theta(1)$
- to propagate, we check each clause
 $\Theta(m)$
- when propagating, each clause where x_i appears is
 $\rightarrow$ eliminated
 $\rightarrow$ converted into a unit clause
- one variable assignment can simplify up to
 m clauses $\rightarrow \Theta(m)$
- each variable must be assigned just once
 $\rightarrow \Theta(n)$

$\therefore \Theta(m + nm)$

checking for contradictions is done in the assignment process.

$$f_0 \quad O(m + mn)$$

↳ each variable is assigned once

↳ one variable simplifies at most m clauses.

FOR YOU: explain why the complexity is not $O(m \cdot mn)$.

while both techniques have the same running time, depending on the formula, one can be more efficient than the other.

2. Consider the following problem:

$$2\text{-COL} = \{G : G \text{ has a 2-colouring}\}$$

- Design an algorithm that decides if a graph has a 2-colouring
- Relate this problem to 2-SAT and conclude that 2-COL is in PTIME.

ALGORITHM

 claim: a graph has a 2-colouring if and only if it's bipartite.

FOR YOU: prove the claim.

The algorithm goes as follow:

while not all nodes are visited:

① pick an unvisited node u , assign color 1

② for every v s.t. $(u,v) \in E$
• assign v with color 2.
→ if v was visited with color 1, reject.

is the algorithm correct?

YES! → it checks for bipartiteness.

② 2-COL IS IN PTIME

the idea here is to "transform" an instance of 2-COL into an instance of 2-SAT.

given $G = (V, E)$

- for $v \in V$, create variable x_v
- for $(u, v) \in E$, create $(x_u \vee x_v)$
and $(\neg x_u \vee \neg x_v)$

claims:

- ① we can construct the formula in polynomial time
- ② φ_G is satisfiable $\iff G$ is 2 colorable
 - color $\sim$ truth assignment

P3.]

(a) determine if φ is satisfiable

$$0. (\neg x_1 \vee x_2) \wedge (\neg x_2 \vee x_3) \wedge (\neg x_3 \vee x_4) \\ \wedge (\neg x_4 \vee x_1) \wedge (\neg x_3) \wedge (x_4)$$

$$x_4 = \text{TRUE}$$

$$x_3 = \text{FALSE}$$

$$1. (\neg x_1 \vee x_2) \wedge (\neg x_2) \wedge (x_1)$$

$$x_2 = \text{FALSE}$$

$$x_1 = \text{TRUE}$$

$$0 \quad \times$$

the formula is not satisfiable

b. Finding a "max" assignment

In the case before, just one clause was not satisfied, so assignment was automatically "maximal"

c. max assignment is

$$x_1 = \text{TRUE}$$

$$x_2 = \text{FALSE}$$

$$x_3 = \text{FALSE}$$

$$x_4 = \text{TRUE}$$

d. a general algorithm

Can we leverage the structure of Horn clauses to create a general algorithm that runs in PTIME?